

Boosting point cloud understanding through graph convolutional network with scale measurement and high-frequency enhancement

Yun Bai^a, Guanlin Li^a, Xuchao Gong^b, Kuijie Zhang^a, Qian Xiao^a, Chaozhi Yang^a, Zongmin Li^{a,*}

^a QINGDAO Institute of Software, College of Computer Science and Technology, China University of Petroleum (East China), Shandong 266580, China

^b Shengli Oilfield Information Technology Service Center, Shandong 257000, China

ARTICLE INFO

Keywords:

Point cloud
Graph convolution
Graph spatial scale
High-frequency information

ABSTRACT

Graph-based methods have exhibited exceptional performance in point cloud understanding by capturing local geometric relationships. However, existing approaches often struggle to characterize the overall spatial scale of local graphs. In addition, they fail to capture the differences between nodes effectively, which is crucial for distinguishing different classes. This study introduces SM-HFEGCN, a novel graph convolutional network that addresses these limitations through two key innovations: scale measurement and high-frequency enhancement. First, we introduce a spatial scale feature derived from the diagonal vectors of the neighborhood, which serves as a unique graph-specific property related to the geometry and density of the local point cloud. This feature can characterize the overall spatial scale of the local point cloud. Second, we enhance the high-frequency information to capture node variations and integrate it with smoothed information to represent the differences and similarities between nodes simultaneously. Extensive experiments demonstrate the effectiveness of SM-HFEGCN in point cloud classification and segmentation tasks.

1. Introduction

Point cloud understanding has emerged as a prominent topic in three-dimensional (3D) vision and has attracted significant attention in recent years. Point clouds represent discrete data distributed in 3D space with disorder and sparsity, posing challenges for capturing effective information. These inherent properties make it difficult to process point clouds directly using standard convolutional neural networks (CNNs). Although some studies have made notable progress in modeling point relationships [1–3], the effective extraction and utilization of these irregular data remain challenging.

Because of their complex topological and geometric properties, the efficient capture of local information within a point cloud is crucial for downstream tasks. Several approaches, such as defining graph convolution operators in the spatial or spectral domains, have been proposed to represent the structured features of local graphs implicitly [1,4–6]. For instance, DGCNN [1] incorporates edge convolution to model the relationship between the centroid and its neighboring points, which are searched using the *k*-nearest neighbor (*KNN*) algorithm. Subsequently, the centroid and neighboring points are connected to construct a graph in local space. DGCNN [1] demonstrates superior performance with max pooling for the graph feature aggregation approach. However, feature aggregation via max pooling only integrates information from the nodes in the neighborhood corresponding to the maximum

value. This limitation implies that the final features can only represent a partial spatial scale of the graph and fail to characterize the overall scale. Although subsequent studies [7,8] have improved DGCNN [1], the issues caused by max pooling persist.

In the spectral domain, graph convolution for point cloud understanding typically follows the graph convolutional network (GCN) [9] and ChebyNet [10] paradigms, in which graph signals are transformed into the spectral domain for feature extraction and then reverted to the spatial domain using inverse Fourier transform. These methods avoid the computational cost of Laplacian matrix decomposition and enhance the efficiency of graph feature extraction. In the spectral domain, low-frequency information reveals the similarities between graph nodes, whereas high-frequency information highlights the differences [11,12]. Most existing studies [1,7,8,13] have focused on the geometric properties of point clouds in the spatial domain; however, capturing the differences between graph nodes in this domain remains challenging. Effective graph feature extraction requires not only capturing the similarities between nodes, but also identifying discriminative features to group them. Because nodes in the same graph undergo multiple feature aggregations, their properties tend to be similar, making it difficult to distinguish them.

* Corresponding author.

E-mail address: lizongmin@upc.edu.cn (Z. Li).

<https://doi.org/10.1016/j.knosys.2024.112715>

Received 2 June 2024; Received in revised form 31 October 2024; Accepted 4 November 2024

Available online 12 November 2024

0950-7051/© 2024 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

We propose two key advancements to overcome these limitations: spatial scale measurement and high-frequency enhancement. The spatial scale feature captures the overall graph structure and is determined by the diagonal vector in the neighborhood space. This feature reflects the local sparsity, with a larger spatial scale indicating a lower density in the local point cloud with a fixed number of neighbors. Furthermore, we enhance the high-frequency information in the spectral domain to improve the differentiation between nodes. This approach emphasizes the high-frequency components, which are crucial for distinguishing fine-grained details and sharpening object contours. In addition, we design a simple graph convolution framework to capture smooth information between nodes. The features processed by this framework are concatenated with differential information, jointly representing the similarities and differences between nodes.

The main contributions of this study are as follows:

- (1) We propose the graph spatial scale feature, which can represent the overall scale and sparsity of a local point cloud.
- (2) We enhance the high-frequency information in the spectral domain and combine it with smoothed information for a joint representation of the differences and similarities between nodes.
- (3) We conducted extensive experiments on several benchmark datasets to demonstrate the effectiveness of the proposed SM-HFEGCN.

2. Related work

2.1. Deep learning on point clouds

The processing of point clouds using neural networks has garnered significant attention in recent years. Owing to their sparse and unordered nature, point clouds cannot be processed effectively using standard CNNs. Several approaches have been proposed to address this issue. Some methods [14–17] involve voxelizing or projecting the point cloud, followed by processing using standard convolution layers. Voxel-based methods [14,15] convert point clouds into voxels and extract features via 3D convolution. Although voxels are more compatible with standard convolution than point clouds, the voxelization process can result in information loss, making it challenging to capture fine-grained features. Multi-view-based methods [16,17] project the point cloud onto a plane and subsequently process it using standard convolution. Although these multi-view-based approaches achieve excellent performance, they incur substantial computational costs.

PointNet [2] and PointNet++ [7] perform feature extraction directly from the original point cloud to avoid such complex processing steps. PointNet [2] constructs a feature extraction network using multilayer perceptrons (MLPs) to capture the features of individual points. PointNet++ [7] builds on PointNet by introducing a set abstraction method to capture local information within the point cloud. Subsequent methods have followed the approaches of PointNet [2] and PointNet++ [7] to explore the geometric properties directly on the original point cloud further. Convolution-based methods [18–23] include convolution operators that conform to the shape of the point cloud, enabling a more effective extraction of shape information while mitigating the effects of sparsity and disorder. Graph-based methods [1,4–6,13,24] model the point cloud as a graph structure and use graph convolution operators to extract local information. Other methods [25,26] focus on modeling the relationships between points and designing shape descriptors to capture the local shape information. To reduce the number of parameters, Point-NN [27] employs a parameter-free learning approach for local point cloud modeling, demonstrating superior performance. In addition, DELA [28] decouples the local information extraction module, significantly reducing the computational requirements. Self-supervised methods [29,30] supplement point clouds and apply pretrained models to point cloud understanding tasks, thereby reducing the manual labeling workload. Inspired by natural language processing, Transformer-based methods [31–35] apply the Transformer architecture to point cloud understanding, yielding excellent performance.

2.2. Graph convolution on point clouds

Graph convolution methods for point cloud understanding can be categorized into two main types: spatial and spectral methods.

Spatial methods. DGCNN [1] is one of the most representative spatial graph convolution approaches for point cloud understanding. DGCNN [1] constructs local directed graphs, explores the effect of aggregation functions on the aggregation of neighboring information, and introduces edge convolution. 3DGCN [13] designs convolution kernels that align with the graph structure, effectively representing its properties. GACNet [8] and AdaptConv [7] enhance edge convolution by assigning edge weights to aggregate node features. GSNet [36] investigates different graph representations in both the spatial and spectral domains, facilitating information interaction across domains with strong scalability. DeepGCNs [5] address the oversmoothing issue in graph convolution and mitigate it by deepening or widening the convolution layers.

Spectral methods. Graph convolution methods in the spectral domain for point cloud understanding stem primarily from the GCN [9] and ChebyNet [10]. RGCNN [4] dynamically updates the Laplacian matrix and performs feature extraction following the GCN [9] framework. LocalSpecGCN [37] introduces recursive clustering and a novel pooling method, replacing traditional max pooling to better preserve graph information. HGNN [38] achieves superior performance by constructing hypergraphs and employing hyperedge convolution. PointGCN [39] and PointNGCNN [40] build on ChebyNet [10], using Chebyshev polynomials as graph filters for feature extraction. PointWavelet [6] includes learnable wavelet transforms within a Transformer framework to capture the contextual information of the point cloud.

Although graph convolution methods have exhibited significant efficacy in point cloud understanding, they are subject to certain limitations. The sparsity of local point clouds is intrinsically linked to their spatial scale, and existing approaches frequently struggle to represent this overall spatial scale accurately. Furthermore, the differences between the graph nodes are critical for distinguishing the points of various classes. However, many current methods prioritize node similarities and neglect these vital distinctions. In this study, we propose a GCN for point cloud understanding that measures the overall spatial scale of the graph and effectively captures both the similarities and differences between nodes.

3. Method

3.1. Simple graph convolution framework

For the input point cloud $P \in \mathbb{R}^{N \times 3}$, where N represents the number of points, the graph $G = (V, E)$ is constructed by connecting each point to its neighbors, which are identified using the KNN algorithm. Here, V denotes the nodes and E represents the edges. p_i denotes a point in point cloud P and $f(p_i)$ is the feature of this point.

The structure of a point cloud in its original space is important. To capture this, we compute the Euclidean distances between the graph nodes in the original space to derive the weight matrix W . The i th row of the weight matrix W indicates the connectivity of graph G_i , where p_i serves as the centroid. The value of W_{ij} is inversely proportional to the distance between points p_i and p_j .

$$W_{ij} = \begin{cases} \frac{1}{\|x_i - x_j\|_2 + 1}, & p_j \in \mathcal{N}_i \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

where x_i denotes the coordinates of point p_i and $\mathcal{N}_i$ denotes the set of neighboring points of p_i .

Following the above operation, the distance between point p_i and its neighboring point $p_j \in \mathcal{N}_i$ can be normalized to $(0,1]$. The weight

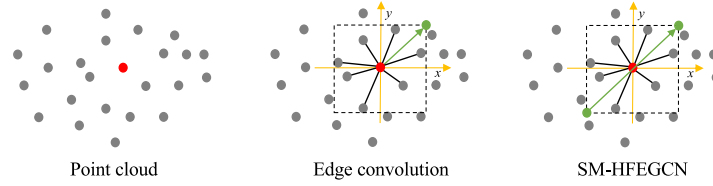


Fig. 1. Difference between SM-HFEGCN and edge convolution in representing the spatial scale of a graph.

matrix W_i is then normalized to represent the weights of the nodes in G_i :

$$\widetilde{W}_{ij} = \frac{W_{ij}}{\sum_j W_{ij}}. \quad (2)$$

The graph feature aggregation process can be expressed as

$$\text{GraphConv}(G_i) = \sum_{j \in \mathcal{N}_i} (\widetilde{W}_{ij}, f(p_j)) \theta_{ij}, \quad (3)$$

where $f(p_j)$ denotes the feature of point p_j and θ_{ij} is the learnable parameter.

As the number of graph convolution layers increases, the receptive field of each point expands, leading to greater overlap among neighboring points. A high neighborhood overlap results in an increased similarity between the node features, causing the graph to become smoother. This similarity can represent the overall properties of the graph.

3.2. Graph spatial scale feature

Owing to the max pooling operation, the edge convolution in DGCNN [1] can represent the partial spatial scale of the graph; specifically, the maximum point within the local space. The following equation expresses the edge convolution:

$$\text{Agg}(G_i) = \text{Max}((f(p_i) - f(p_j), p_j \in \mathcal{N}_i)W), \quad (4)$$

where $f(p_j)$ denotes the feature of point p_j and $\mathcal{N}_i$ denotes the set of neighboring points of p_i .

Although max pooling exhibits strong performance in feature aggregation, it can lead to the loss of neighborhood information. In 3D space, max pooling can be viewed as the selection of a vertex in the cube containing the local graph, where the x , y , and z coordinates of this vertex are maximized within the local space. Although max pooling effectively captures the maximum number of features of graph nodes, it does not fully represent the overall spatial scale of the graph, as illustrated in Fig. 1. In Fig. 1, the red point denotes the center point, with its neighboring points identified using the KNN algorithm. A local graph is formed by connecting the centroid to the neighboring points. Max pooling characterizes the partial spatial scale of the graph, as indicated by the green point in Fig. 1. In contrast, our spatial scale measurement method is designed to capture the overall spatial scale of the neighborhood.

In SM-HFEGCN, the overall spatial scale of the graph is characterized by the graph spatial scale feature ($GSSF$), which is computed using the diagonal vector in the local space.

$$GSSF_i = (\text{Max}(f(p_i) - f(p_j)) - \text{Min}(f(p_i) - f(p_j)), p_j \in \mathcal{N}_i), \quad (5)$$

where $GSSF_i$ denotes the diagonal vector in the space where graph G_i is located and characterizes the overall scale of G_i . Given a fixed number of neighboring points, $GSSF_i$ indicates the neighborhood sparsity. Specifically, a larger $\|GSSF_i\|_2$ indicates a larger neighborhood space, suggesting that the local point cloud is sparser.

3.3. High-frequency enhancement

In the process outlined in Section 3.1, after multiple feature aggregations, the nodes become smooth, leading to their information becoming increasingly similar. However, for point cloud understanding, it is crucial to capture similarities between neighboring points and highlight the differences that characterize variations in the local structure of the point cloud. These differences between nodes can assist in distinguishing and grouping the points of different classes within the same graph, especially at boundaries where multiple classes converge. In the spectral domain, the high-frequency components of the graph reveal critical differences between nodes [11,12]. Therefore, we enhance the high-frequency information of the graph to emphasize the differences between nodes.

For graph G , A is the corresponding adjacency matrix. In the spectral domain, the Laplace matrix L of the graph can be defined as

$$L = I - D^{-1/2} A D^{-1/2}, \quad (6)$$

where I denotes the unit matrix and D is the degree matrix computed through A . L can be orthogonally decomposed as $L = U \Lambda U^T$, where each column u_i of $U \in \mathbb{R}^{N \times N}$ denotes an eigenvector of L and $\lambda_i = \Lambda_{ii}$ denotes the corresponding eigenvalue of u_i .

The convolution in the spectral domain can be defined as

$$x *_G f = U((U^T g) \odot (U^T f)), \quad (7)$$

where $*_G$ is the graph convolution; $\hat{f} = U^T f$ denotes the Fourier transform, which transforms the signal f into the spectral domain; $f = U \hat{f}$ denotes the inverse Fourier transform; g denotes the convolution kernel in the spatial domain, and we use $g_\theta = U^T g$ to denote the convolution kernel in the spectral domain for ease of representation; and $\odot$ denotes the Hadamard product.

We replace $\Lambda_e = (\text{diag}\{e^{\lambda_i}\}_{i=1}^N)$ with $\Lambda = \text{diag}\{\{\lambda_i\}_{i=1}^N\}$ to enhance the high-frequency information of the graph.

Using Taylor polynomials, Λ_e can be defined as

$$\Lambda_e \approx I + \Lambda + \frac{\Lambda^2}{2!} + \dots + \frac{\Lambda^n}{n!}. \quad (8)$$

The Laplace matrix at this point can be expressed as

$$\begin{aligned} L_e &= U \Lambda_e U^T \\ &\approx U \left(I + \Lambda + \frac{\Lambda^2}{2!} + \dots + \frac{\Lambda^n}{n!} \right) U^T \\ &= U U^T + U \Lambda U^T + U \frac{\Lambda^2}{2!} U^T + \dots + U \frac{\Lambda^n}{n!} U^T \\ &= I + L + \frac{L^2}{2!} + \dots + \frac{L^n}{n!} \\ &\approx e^L. \end{aligned} \quad (9)$$

Because U is an orthogonal matrix, $U^T = U^{-1}$, $U U^T = I$, and $U^T U = I$.

The authors of [10] proposed that g_θ can be thought of as a function of the eigenvalues. The convolution kernel g_θ in the spectral domain is defined as follows:

$$g_\theta(\Lambda) = \sum_{k=0}^K \theta_k T_k(\Lambda), \quad (10)$$

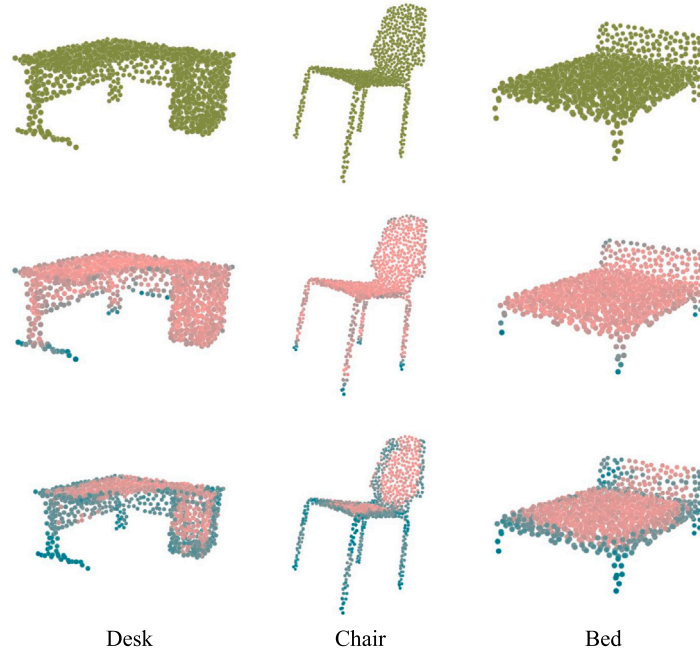


Fig. 2. Visualization of features following graph convolution and *HFE*. The first row shows the original point clouds; the second row shows the features after graph convolution; the final row shows the features following *HFE*.

where K denotes the hyperparameter, θ_k is the polynomial coefficient, and $T_k(x)$ is the Chebyshev polynomial.

After enhancing the high-frequency information, the convolution kernel in the spectral domain can be expressed as

$$\begin{aligned} U g_\theta U^T f &= U \left(\sum_{k=0}^K \theta_k (\Lambda_e)^k \right) U^T f \\ &= \theta_0 U U^T f + \theta_1 (U \Lambda_e U^T) f + \theta_2 (U \Lambda_e^2 U^T) f \\ &\quad + \dots + \theta_K (U \Lambda_e^K U^T) f \\ &= (\theta_0 I + \theta_1 e^L + \theta_2 e^{2L} + \dots + \theta_K e^{KL}) f. \end{aligned} \quad (11)$$

To reduce these parameters, we consider only the first two items, and share the parameters between θ_0 and θ_1 . The high-frequency enhancement (*HFE*) process can be expressed as

$$HFE(f) = \theta (I + e^L) f, \quad (12)$$

where θ is the learnable parameter.

Fig. 2 shows the features obtained after graph convolution and *HFE*. Following graph convolution, the convergence of the point colors reflects the similarity in their features, leading to increased graph smoothness. This smooth propagation of information across nodes is a critical advantage of graph convolution because it promotes a comprehensive understanding of the overall properties of the graph. However, for effective node classification in point cloud understanding, accounting for both the smoothness of the graph and the differences between nodes is essential. Following *HFE*, the edge points appear darker, signifying pronounced property differences.

4. Experiments

In this section, we demonstrate the performance of SM-HFEGCN across multiple benchmark datasets and provide visualizations of the partial experimental results. Extensive ablation studies were conducted to demonstrate the effectiveness of each component of SM-HFEGCN. Thereafter, we present a complexity analysis. Finally, a case study is described, further illustrating the practical application and advantages of SM-HFEGCN in specific scenarios. Fig. 3 presents an overview of SM-HFE Graph Conv, whereas Fig. 4 presents the model architectures employed in this study.

4.1. Experimental settings

4.1.1. Implementation details

SM-HFEGCN was trained and tested on an RTX 4090. We employed the AdamW optimizer and cosine annealing strategy for all tasks, with the weight decay set to 0.05. In the classification task, we set the initial learning rate to 0.002 with 30 neighboring points for ModelNet40 and 20 neighboring points for ScanObjectNN. The ModelNet40 dataset was trained for 500 epochs with a batch size of 32, whereas the ScanObjectNN dataset was trained for 250 epochs with a batch size of 32. For the part segmentation task, the initial learning rate was set to 0.002 with 45 neighboring points, and the model was trained for 250 epochs with a batch size of 16. Finally, in the semantic segmentation task, the initial learning rate was set to 0.006 with 25 neighboring points, and the model was trained for 100 epochs with a batch size of 8.

4.1.2. Benchmarks

ModelNet40. ModelNet40 [41] contains 40 classes of objects sampled from CAD models, among which 9843 models are training data and 2468 models are testing data.

ScanObjectNN. ScanObjectNN [42] is a real scene dataset containing 15 categories with approximately 15,000 objects.

ShapeNetPart. ShapeNetPart [43] contains 16,880 models with 16 categories and 50 parts. The number of parts in each category ranges from two to six.

S3DIS. S3DIS [44] is a large-scale indoor point cloud dataset containing 13 categories. Point-cloud data were collected from six indoor areas with 271 rooms and over 215 million points.

4.2. Classification performance

We present the performance of SM-HFEGCN in the point cloud classification task on both the ModelNet40 and ScanObjectNN datasets. Table 1 reports the classification performance of SM-HFEGCN on ModelNet40, utilizing the xyz coordinates of 1024 points as model inputs for both training and testing. SM-HFEGCN exhibited excellent performance, achieving a comparable overall accuracy (OA) while outperforming PointMLP [3] in terms of the mean class accuracy (mAcc).

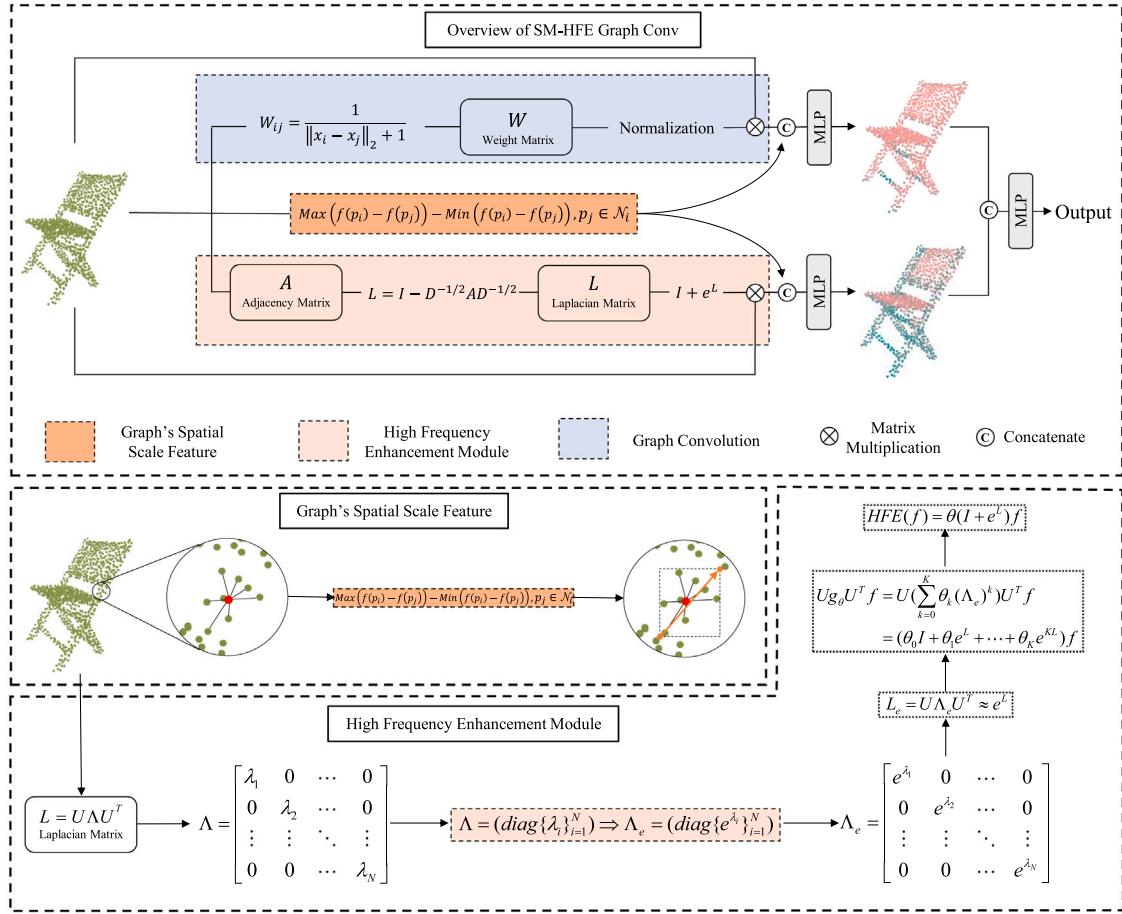


Fig. 3. Illustration of SM-HFE Graph Conv. From top to bottom: an overview of SM-HFE Graph Conv, the GSSF, and the HFE module.

Table 1

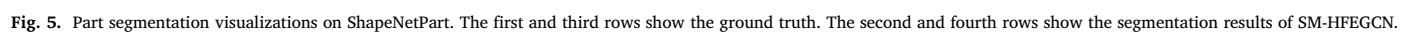
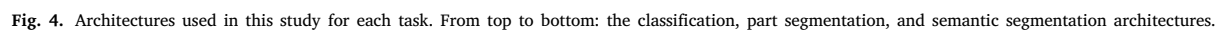
Classification performance on ModelNet40. The top half of the table compares the accuracy of SM-HFEGCN with state-of-the-art methods; the bottom half compares the accuracy of SM-HFEGCN with other GCNs.

Method	Input	OA (%)	mAcc (%)
PointNet [2]	1K points	89.2	86.0
PointNet++ [7]	1K points+normal	91.9	—
PointConv [18]	1K points+normal	92.5	—
RSCNN [45]	1K points	92.9	—
KPConv [21]	7K points	92.9	—
PCT [32]	1K points	93.2	—
PT [31]	1K points	93.7	90.6
PointASNL [46]	1K points	92.9	—
GBNet [47]	1K points	93.8	91.0
GDANet [48]	1K points	93.4	—
PACConv [22]	1K points	93.6	—
CurveNet [25]	1K points	93.8	—
RepSurf-U [26]	1K points	94.4	91.4
PointMLP [3]	1K points	94.1	91.3
PointCont [34]	1K points	93.5	—
MKConv [23]	1K points	93.7	—
SM-HFEGCN	1K points	94.1	91.5
Graph Conv	DGCNN [1]	1K points	92.9
	RGCNN [4]	1K points+normal	90.5
	LocalSpecGCN [37]	2K points+normal	92.1
	3DGCN [13]	1K points	92.1
	PointNGCNN [40]	1K points	92.8
	GSNet [36]	1K points	92.9
	AdaptConv [49]	1K points	93.4
	PointWavelet [47]	1K points	94.1
	SM-HFEGCN	1K points	94.1

PointMLP [3] simplifies the local characterization by designing geometric affine modules and increasing the number of layers in the model. Although PointMLP excelled in local feature extraction, it failed to capture the differences between local points as effectively as SM-HFEGCN, which explains the superior performance of the proposed method. Although SM-HFEGCN trailed RepSurf-U [26] slightly in the OA, it exhibited a superior mAcc. In addition, compared to RepSurf-U [26] and CurveNet [25], our method does not require the construction of approximate surfaces and curves, thereby saving substantial computational resources. Notably, SM-HFEGCN surpassed all other listed graph convolution methods in terms of classification performance. Classical graph convolution methods such as DGCNN [1] and its derivatives GSNet [36] and AdaptConv [49] capture only part of the spatial scale of the graph, whereas SM-HFEGCN captures the overall spatial scale, leading to significantly better classification performance. Although ModelNet40 is a classic dataset for point cloud classification, its applicability to real-world scenes may be limited; therefore, we further evaluated SM-HFEGCN on the real-world dataset ScanObjectNN using 1024 points and their xyz coordinates as model inputs. SM-HFEGCN outperformed all listed methods in the OA and mAcc, as shown in Table 2. Compared to DGCNN [1] and RepSurf-U [26], SM-HFEGCN achieved improvements of 10.6% and 2.7%, respectively, demonstrating the effectiveness of our method.

4.3. Part segmentation performance

Table 3 presents the performance of SM-HFEGCN in the part segmentation task compared with other methods. The model inputs consisted of the coordinates and normal vectors of 2048 points. Because of the ability of SM-HFEGCN to capture the overall spatial scale of



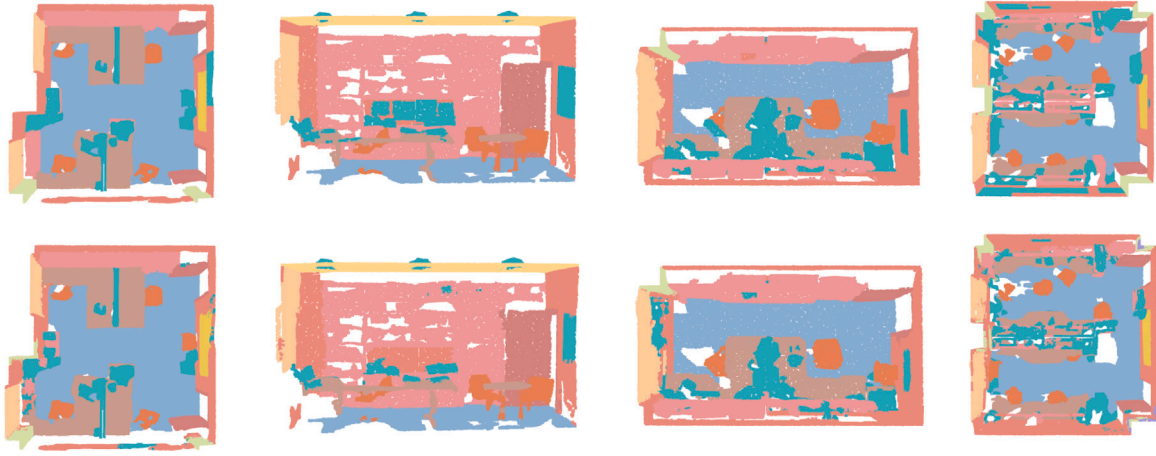


Fig. 6. Semantic segmentation results on S3DIS. The first row shows the ground truth. The second row shows the segmentation results of SM-HFEGCN.

Table 2

Classification performance on ScanObjectNN.

Method	OA (%)	mAcc (%)
PointNet [2]	68.2	63.4
SpiderCNN [19]	73.7	69.8
PointNet++ [7]	77.9	75.4
DGCNN [1]	78.1	73.6
PointCNN [20]	78.5	75.1
GBNet [47]	80.5	77.8
PointMLP [3]	85.4	83.9
RepSurf-U [26]	86.0	–
SPoTr [35]	88.6	86.8
PointConT [34]	88.0	86.0
PointWavelet [6]	87.9	85.5
SM-HFEGCN	88.7	87.0

Table 3

Part segmentation performance on ShapeNetPart.

Method	Inst. mIoU (%)	Cls. mIoU (%)
PointNet [2]	83.7	80.4
PointNet++ [7]	85.1	81.9
RGCNN [4]	84.3	–
PointConv [18]	85.7	82.8
KPConv [21]	86.4	85.1
DGCNN [1]	85.2	82.3
RSCNN [45]	86.2	84.0
AdaptConv [49]	86.4	83.4
GSNet [36]	85.3	–
3DGCN [13]	85.1	82.1
PointMLP [3]	86.1	84.6
PointWavelet [6]	86.8	85.0
SM-HFEGCN	87.0	85.1

the graph, unlike DGCNN [1] and GSNet [36], which only capture partial spatial scales, our method performed better across all metrics. Moreover, by capturing the overall spatial scale of the graph and the differences between graph nodes, SM-HFEGCN exhibited superior performance in classifying local edge points. In addition, SM-HFEGCN exhibited enhanced performance compared with PointWavelet [6], which employs wavelet transforms to process point cloud data in the frequency domain but fails to capture high-frequency information. The inability of PointWavelet to discern the differences between graph nodes effectively highlights the advantage of our approach, which successfully processes both the similarities and differences between nodes to better capture the spatial relationships between them. Visualizations of the part segmentation results are shown in Fig. 5.

4.4. Semantic segmentation performance

We evaluated the performance of SM-HFEGCN in the semantic segmentation task using the large-scale indoor scene dataset S3DIS. SM-HFEGCN employs a data sampling method similar to that of AdaptConv [49], which uniformly samples raw data and is evaluated in Area 5. The model considers the coordinates and colors of the points as inputs. Table 4 presents the performance of SM-HFEGCN in this task, showing that it outperformed the classical semantic segmentation method KPConv [21] in the mIoU, thereby demonstrating superior segmentation across various object categories. Although KPConv [21] introduces deformable convolutions designed for point cloud data, unlike graph convolution methods, it is limited in its ability to extract the structural information of local point clouds. Furthermore, although AdaptConv [49] improves upon DGCNN [1], it still faces challenges related to max pooling and struggles to capture the differences between graph nodes, which makes the accurate grouping of boundary points difficult. In contrast, SM-HFEGCN addresses these limitations by capturing the overall spatial scale of the local point clouds and obtaining the differences between graph nodes, leading to more consistent and accurate segmentation outcomes. Visualizations of some of the semantic segmentation results are presented in Fig. 6, further illustrating the effectiveness of our method.

4.5. Ablation studies

Ablation studies were conducted on ScanObjectNN to assess the importance of each component of SM-HFEGCN. Table 5 lists the effects of each component. The proposed graph convolution method achieved favorable performance on ScanObjectNN, with a notable 6.1% improvement observed after incorporating spatial scale features, thereby demonstrating the effectiveness of the *GSSF*. In addition, the *HFE* component showed a significant improvement of 2.4%. Different spatial scale representations are compared in Table 6, with the proposed spatial scale representation yielding a substantial improvement over max pooling. Furthermore, Table 7 presents the effect of varying the number of neighbors on the classification performance of SM-HFEGCN.

4.6. Complexity analysis

We analyzed the parameters (Param) and floating-point operations (FLOPs) of SM-HFEGCN, as shown in Table 8. Both Param and FLOPs were computed on a single RTX 4090 using 1024 points as the input to the classification architecture. Compared with PointMLP [3],

Table 4
Semantic segmentation performance on S3DIS.

Method	OA	mAcc	mIoU	ceiling	floor	wall	beam	col	wind.	door	table	chair	sofa	book.	board	clut.
PointNet [2]	–	49.0	41.1	88.8	97.3	69.8	0.1	3.9	46.3	10.8	59.0	52.6	5.9	40.3	26.4	33.2
PointCNN [20]	85.9	63.9	57.3	92.3	98.2	79.4	0.0	17.6	22.8	62.1	74.4	80.6	31.7	66.7	62.1	56.7
KPConv [21]	–	72.8	67.1	92.8	97.3	82.4	0.0	23.9	58.0	69.0	81.5	91.0	75.4	75.3	66.7	58.9
AdaptConv [49]	90.0	73.2	67.9	93.9	98.4	82.2	0.0	23.9	59.1	71.3	91.5	81.2	75.5	74.9	72.1	58.6
PointNeXt [50]	90.7	77.5	70.8	94.2	98.5	84.4	0.0	37.7	59.3	74.0	83.1	91.6	77.4	77.2	78.8	60.6
MKConv [23]	89.6	75.1	67.7	92.4	98.2	83.9	0.0	28.5	64.5	65.7	89.7	82.4	73.9	67.5	77.3	55.9
SM-HFEGCN	91.8	78.0	71.9	95.6	98.7	86.7	0.0	40.0	60.9	71.9	85.2	92.5	79.6	80.8	77.8	65.2

Table 5
Ablation study of different components.

Graph Conv	GSSF	HFE	OA (%)
✓			80.2
✓	✓		86.3
✓		✓	85.2
✓	✓	✓	88.7

Table 6
Ablation study of spatial scale measurement strategies.

GSSF	Max	Min	Max-min
OA (%)	86.3	86.7	88.7

Table 7
Ablation study of neighbor numbers.

k (number of neighboring points)	15	20	25
OA (%)	88.2	88.7	87.9

Table 8
Complexity analysis of SM-HFEGCN in classification.

Method	Param (M)	FLOPs (G)
PointNet [2]	3.47	0.45
DGCNN [1]	2.61	2.95
CurveNet [25]	2.14	0.32
PointMLP [3]	13.2	15.7
SM-HFEGCN	5.88	0.89

SM-HFEGCN demonstrated a clear advantage in both metrics. Compared with DGCNN [1], SM-HFEGCN had a higher number of parameters but fewer FLOPs. This difference may be attributed to the additional parameters required by SM-HFEGCN for learning in the spectral domain.

4.7. Case studies

4.7.1. Classification case study

We present the performance of SM-HFEGCN on the ModelNet40 dataset across various classes, as detailed in Table 9, and provide a comparative analysis with the graph convolution methods DGCNN [1] and AdaptConv [49]. A total of 14 representative classes are listed to provide a focused comparison across different object types. As shown in Table 9, SM-HFEGCN outperformed the other methods, particularly for objects characterized by simple edge contours. To substantiate the effectiveness of SM-HFEGCN further, we selected three representative classes—bathtubs, cones, and cups—for feature map visualization. Fig. 7 illustrates the feature maps corresponding to eigenvalues λ and e^A . As observed in Fig. 7, although the features corresponding to

eigenvalue λ could highlight certain edge points, these points alone were insufficient to represent the entire outline of the object. This could easily lead to confusion among objects from different classes that share similar edge features. However, after applying *HFE*, more edge points were highlighted, allowing for a more accurate and distinct representation of the contour of the object.

Although SM-HFEGCN outperformed DGCNN [1] and AdaptConv [49] in the flower pot class, it achieved an accuracy of only 40%. Consequently, we selected the flower pot class to conduct a more detailed comparison among SM-HFEGCN, DGCNN [1], and AdaptConv [49]. Fig. 8 presents the feature maps generated after feature extraction by the SM-HFE graph convolution in SM-HFEGCN, edge convolution in DGCNN [1], and convolution layer in AdaptConv [49]. Following DGCNN [1] processing, the differences between the edges and interior points of the flower pot decreased, particularly in the region beneath the pot. AdaptConv [49] encountered similar problems. However, following processing with the SM-HFE graph convolution, the differences between the edges and interior points of the flower pot became significantly more pronounced, as highlighted by the red box in Fig. 8. This finding confirms that the *HFE* module in SM-HFEGCN effectively captures these differences and emphasizes the contours of the object.

The classification architecture utilized in this study incorporates multiple downsampling stages. In flower pots, repeated downsampling reduces the number of retained points, which can lead to inaccuracies in the final classification. This limitation contributes to the 40% accuracy of SM-HFEGCN in the flower pot class. Our future work will focus on addressing this challenge, specifically by developing strategies to minimize information loss during downsampling.

4.7.2. Part segmentation case study

To validate the effectiveness of SM-HFEGCN further, we selected two representative classes—airplane and chair—for analysis and comparison with other methods in the part segmentation task. Fig. 9 illustrates the segmentation performance of SM-HFEGCN for these classes compared with the results from DGCNN [1] and AdaptConv [49]. SM-HFEGCN performed better in segmenting the airplane, significantly surpassing DGCNN [1] and AdaptConv [49] in classifying boundary points. Furthermore, SM-HFEGCN excelled in recognizing and segmenting small objects on the airplane, achieving results that exceeded the capabilities of other methods. For chair segmentation, SM-HFEGCN notably outperformed the other methods in accurately classifying the boundary points between the armrests and chair body, which were closely aligned with the ground truth. These findings, as highlighted by the red boxes in Fig. 9, underscore the superiority of SM-HFEGCN in boundary point classification and its effectiveness in capturing node differences, which enhances the ability to group points of different classes at the boundaries.

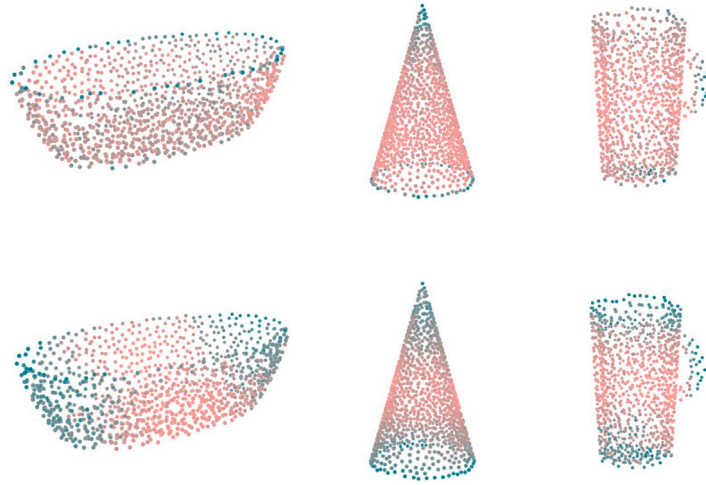
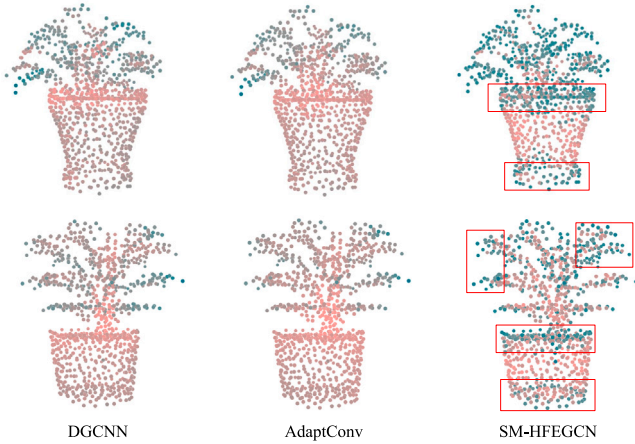
5. Conclusion

This study has proposed SM-HFEGCN, which is a novel graph convolution method for point cloud understanding that incorporates scale

Table 9

Performance of SM-HFEGCN for each class on the ModelNet40 dataset, focusing on 14 representative classes.

Method	OA	mAcc	airp.	bath.	bed	bench	book.	bottle	bowl	chair	cone	cup	curt.	desk	door	flow.
DGCNN [1]	92.9	90.2	100.0	90.0	99.0	80.0	99.0	97.0	95.0	98.0	95.0	45.0	95.0	89.5	95.0	20.0
AdaptConv [49]	93.4	90.7	100.0	96.0	99.0	70.0	100.0	98.0	95.0	99.0	95.0	50.0	95.0	94.2	85.0	20.0
SM-HFEGCN	94.1	91.5	100.0	98.0	100.0	75.0	99.0	98.0	100.0	99.0	100.0	60.0	90.0	93.0	95.0	40.0

**Fig. 7.** Visualization of features with eigenvalues 1 and 2. The first row displays features with eigenvalue λ , while the second row depicts features with eigenvalue e^λ .**Fig. 8.** Features after graph convolution layer. The first column presents the features following edge convolution in DGCNN, the second column shows the features after the convolution layer in AdaptConv, and the third column displays the features after SM-HFE graph convolution in SM-HFEGCN.

measurement and high-frequency enhancement. A central challenge in existing graph convolution methods is their limited ability to represent the overall spatial scale of the graph and their tendency to overlook the differences between graph nodes. SM-HFEGCN addresses these issues using two key components: the *GSSF* and *HFE* module. The *GSSF* captures the overall spatial scale of the graph by computing the diagonal vectors within the local space in which the graph resides. The *HFE* module enhances high-frequency information in the spectral domain

and integrates it with smoothed information to represent the differences and similarities among nodes. Extensive experiments demonstrated that SM-HFEGCN delivers outstanding performance across multiple benchmark datasets.

6. Discussion

Strengths. The proposed SM-HFEGCN has two important components: the *GSSF* and *HFE* module. The *GSSF* captures the overall spatial scale of the graph to represent the spatial nature and sparsity of the local point cloud effectively, which is crucial for analyzing the distribution of the local points. The *HFE* module captures the high-frequency information of the graph to highlight the differences between nodes. This enhancement sharpens the boundaries, which is beneficial for classifying points at different locations. These components enable a more comprehensive representation of the local point cloud and improve the performance of the point cloud classification and segmentation tasks.

Limitations. Although SM-HFEGCN exhibits excellent performance in point cloud understanding tasks, it has average performance in terms of Param and FLOPs compared with some classical methods. More Param and FLOPs mean that more computational resources will be consumed, which is not favorable for the real-time application of the method.

In addition, the classification network suffers from significant information loss owing to multiple downsampling stages, which disrupts the original structure of the object and adversely affects the final classification results.

Future Directions. In future work, we will focus on efficiently mining the properties of point clouds in both the spatial and spectral domains while reducing unnecessary computations. For the classification task, we aim to design a more effective downsampling strategy that minimizes information loss and preserves crucial information.

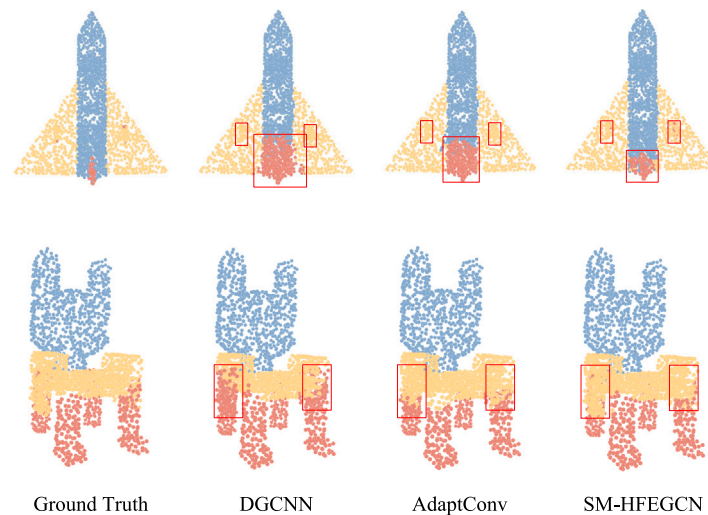


Fig. 9. Comparison of SM-HFEGCN with other methods for the part segmentation task. The first column displays the ground truth, the second column shows the segmentation results of DGCNN, the third column presents the results of AdaptConv, and the fourth column illustrates the results of SM-HFEGCN.

CRedit authorship contribution statement

Yun Bai: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation, Conceptualization. **Guanlin Li:** Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Xuchao Gong:** Visualization, Validation, Software, Project administration, Investigation. **Kuijie Zhang:** Writing – review & editing, Visualization, Validation, Project administration. **Qian Xiao:** Visualization, Validation. **Chaozhi Yang:** Visualization, Validation. **Zongmin Li:** Supervision, Resources, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Key R&D Program (Grant no. 2019YFF0301800), National Natural Science Foundation of China (Grant no. 61379106), and Shandong Provincial Natural Science Foundation, China (Grant nos. ZR2013FM036 and ZR2015FM011).

Data availability

Data will be made available on request.

References

- [1] Y. Wang, Y. Sun, Z. Liu, S.E. Sarma, M.M. Bronstein, J.M. Solomon, Dynamic graph cnn for learning on point clouds, *ACM Trans. Graph. (tog)* 38 (5) (2019) 1–12.
- [2] C.R. Qi, H. Su, K. Mo, L.J. Guibas, Pointnet: Deep learning on point sets for 3d classification and segmentation, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 652–660.
- [3] X. Ma, C. Qin, H. You, H. Ran, Y. Fu, Rethinking network design and local geometry in point cloud: A simple residual MLP framework, 2022, *arXiv preprint arXiv:2202.07123*.
- [4] G. Te, W. Hu, A. Zheng, Z. Guo, Rgcnn: Regularized graph cnn for point cloud segmentation, in: *Proceedings of the 26th ACM International Conference on Multimedia*, 2018, pp. 746–754.
- [5] G. Li, M. Muller, A. Thabet, B. Ghanem, Deepgcns: Can gcns go as deep as cnns? in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 9267–9276.
- [6] C. Wen, J. Long, B. Yu, D. Tao, PointWavelet: Learning in spectral domain for 3-D point cloud analysis, *IEEE Trans. Neural Netw. Learn. Syst.* (2024).
- [7] C.R. Qi, L. Yi, H. Su, L.J. Guibas, Pointnet++: Deep hierarchical feature learning on point sets in a metric space, *Adv. Neural Inf. Process. Syst.* 30 (2017).
- [8] L. Wang, Y. Huang, Y. Hou, S. Zhang, J. Shan, Graph attention convolution for point cloud semantic segmentation, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 10296–10305.
- [9] T.N. Kipf, M. Welling, Semi-supervised classification with graph convolutional networks, 2016, *arXiv preprint arXiv:1609.02907*.
- [10] M. Defferrard, X. Bresson, P. Vandergheynst, Convolutional neural networks on graphs with fast localized spectral filtering, *Adv. Neural Inf. Process. Syst.* 29 (2016).
- [11] D. Bo, X. Wang, C. Shi, H. Shen, Beyond low-frequency information in graph convolutional networks, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 35, 2021, pp. 3950–3957.
- [12] S. Pang, K. Zhang, G. Wang, J.C.-W. Lin, F. Wang, X. Meng, S. Wang, Y. Zhang, AF-GCN: Completing various graph tasks efficiently via adaptive quadratic frequency response function in graph spectral domain, *Inform. Sci.* 623 (2023) 469–480.
- [13] H. Cho, I.S. Choi, Three-dimensionally embedded graph convolutional network (3dgcnn) for molecule interpretation, 2018, *arXiv preprint arXiv:1811.09794*.
- [14] B. Graham, L. Van der Maaten, Submanifold sparse convolutional networks, 2017, *arXiv preprint arXiv:1706.01307*.
- [15] B. Graham, M. Engelcke, L. Van Der Maaten, 3D semantic segmentation with submanifold sparse convolutional networks, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 9224–9232.
- [16] A. Goyal, H. Law, B. Liu, A. Newell, J. Deng, Revisiting point cloud shape classification with a simple and effective baseline, in: *International Conference on Machine Learning*, PMLR, 2021, pp. 3809–3820.
- [17] A. Hamdi, S. Giancola, B. Ghanem, Mvtn: Multi-view transformation network for 3d shape recognition, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 1–11.
- [18] W. Wu, Z. Qi, L. Fuxin, Pointconv: Deep convolutional networks on 3d point clouds, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 9621–9630.
- [19] Y. Xu, T. Fan, M. Xu, L. Zeng, Y. Qiao, Spiderncnn: Deep learning on point sets with parameterized convolutional filters, in: *Proceedings of the European Conference on Computer Vision*, ECCV, 2018, pp. 87–102.
- [20] Y. Li, R. Bu, M. Sun, W. Wu, X. Di, B. Chen, Pointcnn: Convolution on x-transformed points, *Adv. Neural Inf. Process. Syst.* 31 (2018).
- [21] H. Thomas, C.R. Qi, J.-E. Deschaud, F. Marcote, F. Goulette, L.J. Guibas, Kpconv: Flexible and deformable convolution for point clouds, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 6411–6420.
- [22] M. Xu, R. Ding, H. Zhao, X. Qi, Paconv: Position adaptive convolution with dynamic kernel assembling on point clouds, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 3173–3182.
- [23] S. Woo, D. Lee, S. Hwang, W.J. Kim, S. Lee, MKConv: Multidimensional feature representation for point cloud analysis, *Pattern Recognit.* 143 (2023) 109800.
- [24] G. Pan, J. Wang, R. Ying, P. Liu, 3DTi-net: Learn inner transform invariant 3d geometry features using dynamic gcn, 2018, *arXiv preprint arXiv:1812.06254*.
- [25] T. Xiang, C. Zhang, Y. Song, J. Yu, W. Cai, Walk in the cloud: Learning curves for point clouds shape analysis, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 915–924.

- [26] H. Ran, J. Liu, C. Wang, Surface representation for point clouds, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 18942–18952.
- [27] R. Zhang, L. Wang, Z. Guo, Y. Wang, P. Gao, H. Li, J. Shi, Parameter is not all you need: Starting from non-parametric networks for 3d point cloud analysis, 2023, arXiv preprint [arXiv:2303.08134](https://arxiv.org/abs/2303.08134).
- [28] B. Chen, Y. Xia, Y. Zang, C. Wang, J. Li, Decoupled local aggregation for point cloud learning, 2023, arXiv preprint [arXiv:2308.16532](https://arxiv.org/abs/2308.16532).
- [29] Y. Pang, W. Wang, F.E. Tay, W. Liu, Y. Tian, L. Yuan, Masked autoencoders for point cloud self-supervised learning, in: European Conference on Computer Vision, Springer, 2022, pp. 604–621.
- [30] Y. Zha, H. Ji, J. Li, R. Li, T. Dai, B. Chen, Z. Wang, S.-T. Xia, Towards compact 3d representations via point feature enhancement masked autoencoders, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 38, 2024, pp. 6962–6970.
- [31] H. Zhao, L. Jiang, J. Jia, P.H. Torr, V. Koltun, Point transformer, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 16259–16268.
- [32] M.-H. Guo, J.-X. Cai, Z.-N. Liu, T.-J. Mu, R.R. Martin, S.-M. Hu, Pct: Point cloud transformer, *Comput. Vis. Media* 7 (2021) 187–199.
- [33] C. Zhang, H. Wan, X. Shen, Z. Wu, PVT: Point-voxel transformer for point cloud learning, *Int. J. Intell. Syst.* 37 (12) (2022) 11985–12008.
- [34] Y. Liu, B. Tian, Y. Lv, L. Li, F.-Y. Wang, Point cloud classification using content-based transformer via clustering in feature space, *IEEE/CAA J. Autom. Sin.* 11 (1) (2023) 231–239.
- [35] J. Park, S. Lee, S. Kim, Y. Xiong, H.J. Kim, Self-positioning point-based transformer for point cloud understanding, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 21814–21823.
- [36] M. Xu, Z. Zhou, Y. Qiao, Geometry sharing network for 3d point cloud classification and segmentation, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 34, 2020, pp. 12500–12507.
- [37] C. Wang, B. Samari, K. Siddiqi, Local spectral graph convolution for point set feature learning, in: Proceedings of the European Conference on Computer Vision, ECCV, 2018, pp. 52–66.
- [38] Y. Feng, H. You, Z. Zhang, R. Ji, Y. Gao, Hypergraph neural networks, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 33, 2019, pp. 3558–3565.
- [39] Y. Zhang, M. Rabbat, A graph-cnn for 3d point cloud classification, in: 2018 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP, IEEE, 2018, pp. 6279–6283.
- [40] Q. Lu, C. Chen, W. Xie, Y. Luo, PointNGCNN: Deep convolutional networks on 3D point clouds with neighborhood graph filters, *Comput. Graph.* 86 (2020) 42–51.
- [41] Z. Wu, S. Song, A. Khosla, F. Yu, L. Zhang, X. Tang, J. Xiao, 3D shapenets: A deep representation for volumetric shapes, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2015, pp. 1912–1920.
- [42] M.A. Uy, Q.-H. Pham, B.-S. Hua, T. Nguyen, S.-K. Yeung, Revisiting point cloud classification: A new benchmark dataset and classification model on real-world data, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2019, pp. 1588–1597.
- [43] L. Yi, V.G. Kim, D. Ceylan, I.-C. Shen, M. Yan, H. Su, C. Lu, Q. Huang, A. Sheffer, L. Guibas, A scalable active framework for region annotation in 3d shape collections, *ACM Trans. Graph. (tog)* 35 (6) (2016) 1–12.
- [44] I. Armeni, O. Sener, A.R. Zamir, H. Jiang, I. Brilakis, M. Fischer, S. Savarese, 3D semantic parsing of large-scale indoor spaces, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 1534–1543.
- [45] Y. Liu, B. Fan, S. Xiang, C. Pan, Relation-shape convolutional neural network for point cloud analysis, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 8895–8904.
- [46] J.-W. Han, D.-J. Synn, T.-H. Kim, H.-C. Chung, J.-K. Kim, Feature based sampling: a fast and robust sampling method for tasks using 3D point cloud, *IEEE Access* 10 (2022) 58062–58070.
- [47] S. Qiu, S. Anwar, N. Barnes, Geometric back-projection network for point cloud classification, *IEEE Trans. Multimed.* 24 (2021) 1943–1955.
- [48] M. Xu, J. Zhang, Z. Zhou, M. Xu, X. Qi, Y. Qiao, Learning geometry-disentangled representation for complementary understanding of 3d object point cloud, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 35, 2021, pp. 3056–3064.
- [49] H. Zhou, Y. Feng, M. Fang, M. Wei, J. Qin, T. Lu, Adaptive graph convolution for point cloud analysis, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 4965–4974.
- [50] G. Qian, Y. Li, H. Peng, J. Mai, H. Hammoud, M. Elhoseiny, B. Ghanem, Pointnext: Revisiting pointnet++ with improved training and scaling strategies, *Adv. Neural Inf. Process. Syst.* 35 (2022) 23192–23204.